

Year 9 Mathematics

NAPLAN 2015 Non calculator

Total Working: 40 minutes

Equipment Allowed: Pen, Pencil, Eraser

1

This table shows a pattern. The top and bottom numbers are connected by a rule.

Top number	1	2	3	4		?
Bottom number	3	6	9	12		27

What is the top number when the bottom number is 27?

- 5 9 15 19
- ☐ ☐ ☐ ☐
-

2

Luca had a \$10 voucher to spend on games for his smartphone.
He bought three games at \$1.99 each.

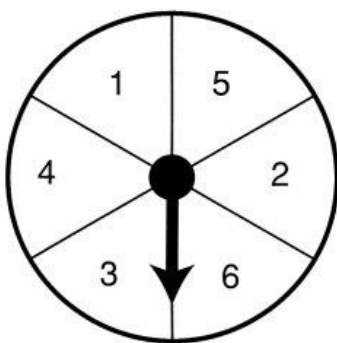
Which expression shows how much he has left to spend?

- ☐ \$10 – \$1.99
- ☐ \$10 – (3 × \$1.99)
- ☐ \$10 – \$1.99 + \$1.99 + \$1.99
- ☐ 3 × (\$10 – \$1.99)
-

3

This spinner is spun twice.

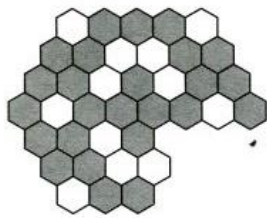
The two numbers that the arrow lands on are added.



Which of these totals is most likely to occur?

- 7 10 11 12
- ☐ ☐ ☐ ☐
-

- 4 Some tiles are missing from this pattern of tiles.



When complete the pattern has two lines of symmetry.
Which of these could be the missing part of the pattern?



☐



☐

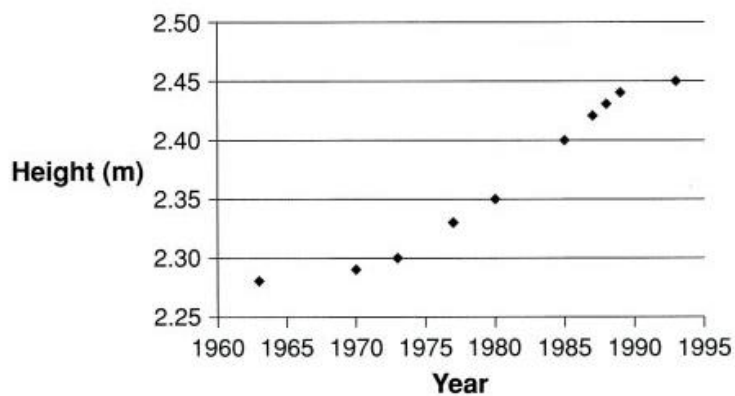


☐



☐

- 5 This graph shows the history of the men's high jump world record.



By about how many centimetres did the world record increase between 1970 and 1985?

9

☐

11

☐

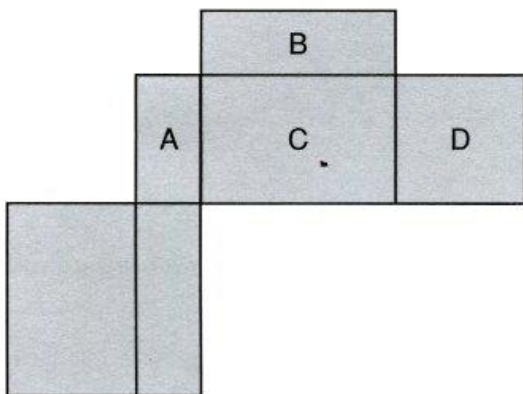
15

☐

17

☐

- 6 Brad was trying to draw a net of a rectangular prism.



He drew one face of the prism incorrectly.
Which face did he draw incorrectly?

A

☐

B

☐

C

☐

D

☐

7 $5 \times 2^3 = ?$

30

☐

40

☐

115

☐

1000

☐

8 $\text{😊} = 6$

$\text{😊} \times \text{☆} = \text{☆} + \text{☆} + \text{☆} + \text{😊}$

What is the value of ☆ ?

0

☐

1

☐

2

☐

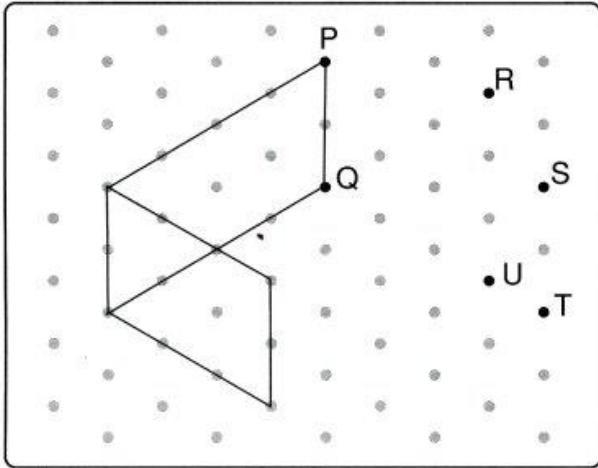
3

☐

4

☐

9 Ann was drawing a rectangular prism on an isometric grid.



Which pair of points should she join with a line?

P and R

☐

P and S

☐

Q and U

☐

Q and T

☐

10



What is the best estimate for the capacity of this mug?

0.30 litres

☐

3.0 litres

☐

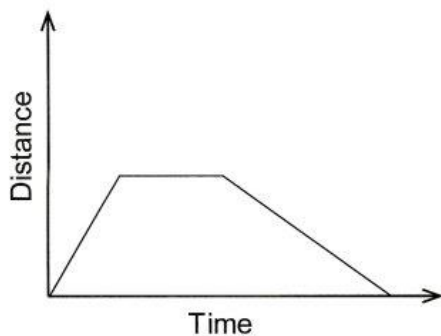
90 millilitres

☐

900 millilitres

☐

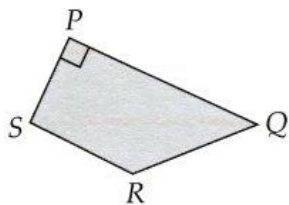
11



Which of the following stories best fits the graph above?

- ☐ Melanie drove north-east, did some shopping and drove back home.
- ☐ Melanie walked uphill, had a rest and ran downhill.
- ☐ Melanie ran to the park, rested for a while and then walked home.
- ☐ Melanie walked to the beach, had a swim and then ran back home.

12



Which of these properties makes $PQRS$ a trapezium?

- ☐ Line PS is perpendicular to line PQ .
- ☐ Line QR is parallel to line SR .
- ☐ Line PR is perpendicular to line SQ .
- ☐ Line PQ is parallel to line SR .

13

Which expression is equal to $5x - 2 + 3x + 6$?

$12x$

☐

$8x + 4$

☐

$8x - 8$

☐

$3x + 9$

☐

14

Ben and Lee were playing a game.

They both started with zero points.

Ben's final score was 240 points. Lee's final score was -60 .

How many more points did Ben have than Lee?

-300

☐

-180

☐

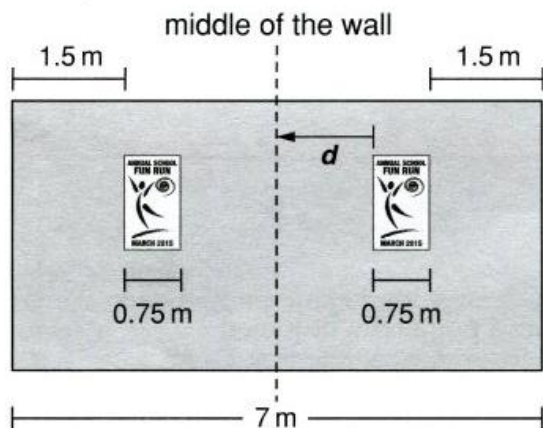
180

☐

300

☐

- 15 Sanjeet put two identical posters on his wall as shown.



What is the distance d from the edge of each poster to the middle of the wall?

- 1.25 m 1.5 m 2.25 m 2 m 4.75 m
- ☐ ☐ ☐ ☐ ☐

- 16 Stuart made a phone call that lasted 59 minutes.

The cost per minute for the phone call was \$0.89.

Stuart estimated that the total cost of the phone call was about \$54.

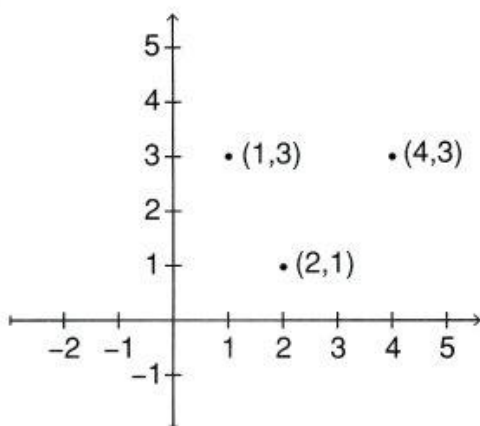
Stuart's estimate was

- ☐ less than the actual cost.
☐ equal to the actual cost.
☐ more than the actual cost.

- 17 Which of these is the closest to 0?

- 1.001 0.01 -0.1 -0.001
- ☐ ☐ ☐ ☐

- 18 Rani draws and labels 3 points, as shown below.



She wants to add another point so that all 4 points can be joined to make a parallelogram.

Which of these is a possible coordinate for the 4th point?

- (-1, 1) (1, 4) (1, -1) (4, 1)
- ☐ ☐ ☐ ☐

19

Nick works in a factory packing chocolates into boxes.

He packs one box at a time.

Each box contains 5 milk chocolates, 3 dark chocolates and 2 white chocolates.

Every hour Nick packs a total of 300 chocolates.

How many dark chocolates does he pack every hour?

20

A standard six-sided dice is rolled once.

What is the probability that the number on the top face is a factor of 6?

$$\frac{1}{6}$$

☐

$$\frac{1}{3}$$

☐

$$\frac{1}{2}$$

☐

$$\frac{2}{3}$$

☐

21

Simplify: $\frac{2^3 \times 5^2 \times 3^4}{3 \times 3^3 \times 5 \times 2^2}$

$$10$$

☐

$$\frac{4}{3}$$

☐

$$1$$

☐

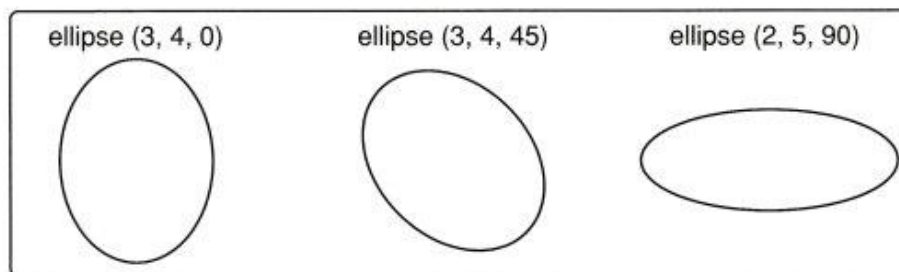
$$\frac{1}{3}$$

☐

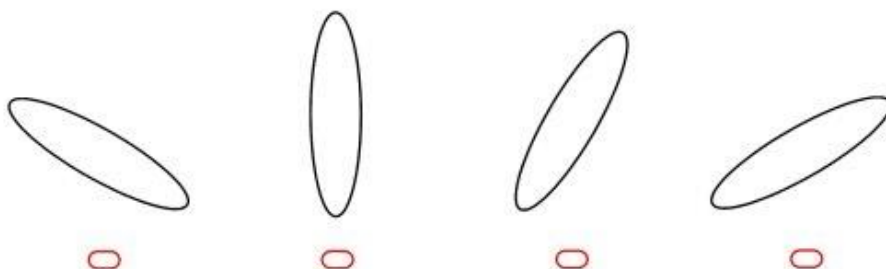
22

Sandra is using a computer program to draw shapes.

The picture shows three commands and the resulting shapes.



Which shape would be drawn by this command: ellipse (1, 4, 60)?



23

It takes 5 hours for 4 people to deliver leaflets to 1000 homes.

At the same rate, how many hours should it take for 5 people to deliver leaflets to 1500 homes?

hours

- 24 Tilly made a number pattern using this rule:

$$\text{next number} = \text{previous number} \times \blacksquare - \triangle$$

($\blacksquare$ and $\triangle$ are whole numbers)

Her number pattern is:

2, 1, -1, -5, -13, ...

What is the next number in Tilly's pattern?

-29

☐

-26

☐

-25

☐

-21

☐

- 25 In a group of Year 9 students, the ratio of boys to girls is 9:7.

There are 4 more boys than girls in this group.

How many students are in this group altogether?

16

☐

18

☐

20

☐

32

☐

- 26 $\frac{x}{2} = \frac{3}{11}$

What is the value of x ?

$\frac{3}{22}$

☐

$\frac{6}{22}$

☐

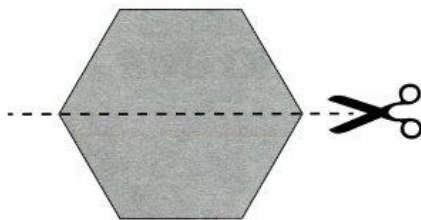
$\frac{5}{11}$

☐

$\frac{6}{11}$

☐

- 27 A regular hexagon has a perimeter of 12 cm. It is then cut in half.



What is the perimeter of each half of the hexagon?

5 cm

☐

6 cm

☐

8 cm

☐

9 cm

☐

10 cm

☐

- 28 Lina drew a line with equation $y = 3 - x$ on a grid.
She then drew another line on the same grid with this equation $y = x - 1$.

What are the coordinates of the intersection point of these two lines?

(-1, 2)

☐

(1, 2)

☐

(2, -1)

☐

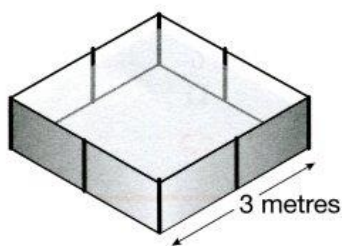
(2, 1)

☐

29

A square pool for small children has sides which are 3 metres long.

One cubic metre can hold 1000 litres of water.



How many litres of water will fill the pool to a depth of 50 centimetres?

litres

30

The area of a rectangle is 60 mm^2 .

The side lengths are doubled to make a new rectangle.

What is the area of the new rectangle?

30 mm^2

☐

120 mm^2

☐

240 mm^2

☐

360 mm^2

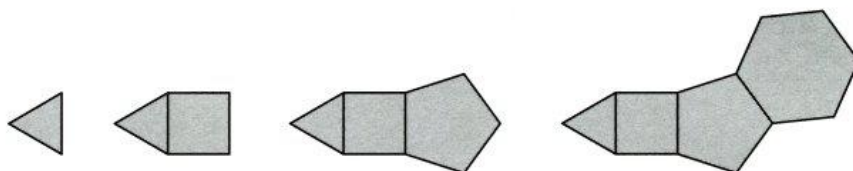
☐

31

Holly created a shape using various regular polygons.

She began by using a triangle with a side length of 1 cm.

She then added regular polygons, each with one more side than the previous one.



She continued this way until a 10-sided polygon is added to her shape.

What will the perimeter of her whole shape be?

cm

32

Kumi is $\frac{3}{4}$ the height of Zac.

Sue is $\frac{2}{3}$ the height of Zac.

Kumi is 15 centimetres taller than Sue.

How tall is Zac in centimetres?

centimetres